# How AI Sees

Name _____

Work with a partner. Compare answers when you finish each problem.

**Word bank:** 0 3 255 blue

**1. Color by number.** Color every square with a **1**. Leave every **0** white.

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 |
| 0 | 1 | 1 | 1 | 1 | 0 | 0 | 1 | 1 | 1 | 1 | 0 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 |
| 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 |
| 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

What is the picture? _____

**2. Pixels are numbers.**

In a black-and-white picture, _____ is black and _____ is white.

In a color picture, each pixel is _____ numbers: red, green, and _____.

### 3. Find the edges on screen.

With the **up-and-down** filter, the bright squares are on the _____

I clicked a bright square. The answer was _____ I clicked a dark square. The answer was _____.

With the **side-to-side** filter, the bright squares are on the _____

A big answer means _____

### 4. Talk about it. What could a self-driving car miss?

A self-driving car could make a mistake when _____

Computer vision is helpful when _____ and dangerous when _____

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**Go Further (Day 2).** Find the edge by hand. Here is a tiny picture (0 = dark, 9 = bright) and a filter.

| | | | | | |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 9 | 9 | 9 |
| 0 | 0 | 0 | 9 | 9 | 9 |
| 0 | 0 | 0 | 9 | 9 | 9 |

**Filter:**

| | | |
|---|---|---|
| −1 | 0 | 1 |
| −1 | 0 | 1 |
| −1 | 0 | 1 |

Put the filter over 3 columns of the picture. Multiply each pair of numbers, then add all 9.

| Filter over columns | Math | Answer |
|---|---|---|
| 1, 2, 3 | | |
| 2, 3, 4 | | |
| 3, 4, 5 | | |
| 4, 5, 6 | | |

Where are the big answers? _____